A DATA VISUALIZATION FINAL PROJECT

The Shifting Landscape of Global Clinical Trials

13,748 trials across 10 major pharmaceutical sponsors, 1984–2020 — what the record of the pre-AI era reveals about where digital health and AI-driven trial design need to go next.

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WHY THIS DATASET

Reading my own industry’s data, not a stranger’s

1 **3 years at IQVIA**

Worked directly with clinical trial operations data — this dataset speaks a language I already read fluently.

2 **A career-defining direction**

Pharma + data science + AI is where clinical development is heading; this project is a deliberate first step, not a class exercise.

3 **A live industry question**

AI and digital health promise faster, cheaper, more targeted trials. The historical record (1984–2020) is the baseline that promise has to beat.

4 **Real, rich, and unresolved**

No cleaned-up teaching dataset — messy sponsors, real terminations, real gaps. Exactly the kind of data a working data scientist inherits.

A rich, multi-dimensional record of real drug development

ClinicalTrials.gov trial registrations, sourced via Kaggle ("A Quick Overview of Clinical Trials")

13,748
trial records

10
major sponsors

1984–2020
years of history

4
trial phases tracked

Attribute types combined in the analysis

Categorical
Sponsor, Phase, Status, Condition

Numerical
Enrollment size

Temporal
Start Year / Month, 36-year span

Text
Trial Title, Summary (condition classification)

Being honest about what the data can and can't say

1

No spatial column

The dataset has no country/site field, so no map-based view was possible. Every other attribute type (categorical, numerical, temporal, text) is combined instead — the brief asks to combine where possible, not force all four.

2

Zero-enrollment trials excluded

Records with Enrollment = 0 (never actually enrolled) are dropped from every enrollment-based analysis to avoid distorting medians and distributions.

3

Tiny-sample decades excluded

The 1980s (n=1) and 2020s (n=2) are excluded from phase-mix-by-decade comparisons — too few trials to support a percentage breakdown.

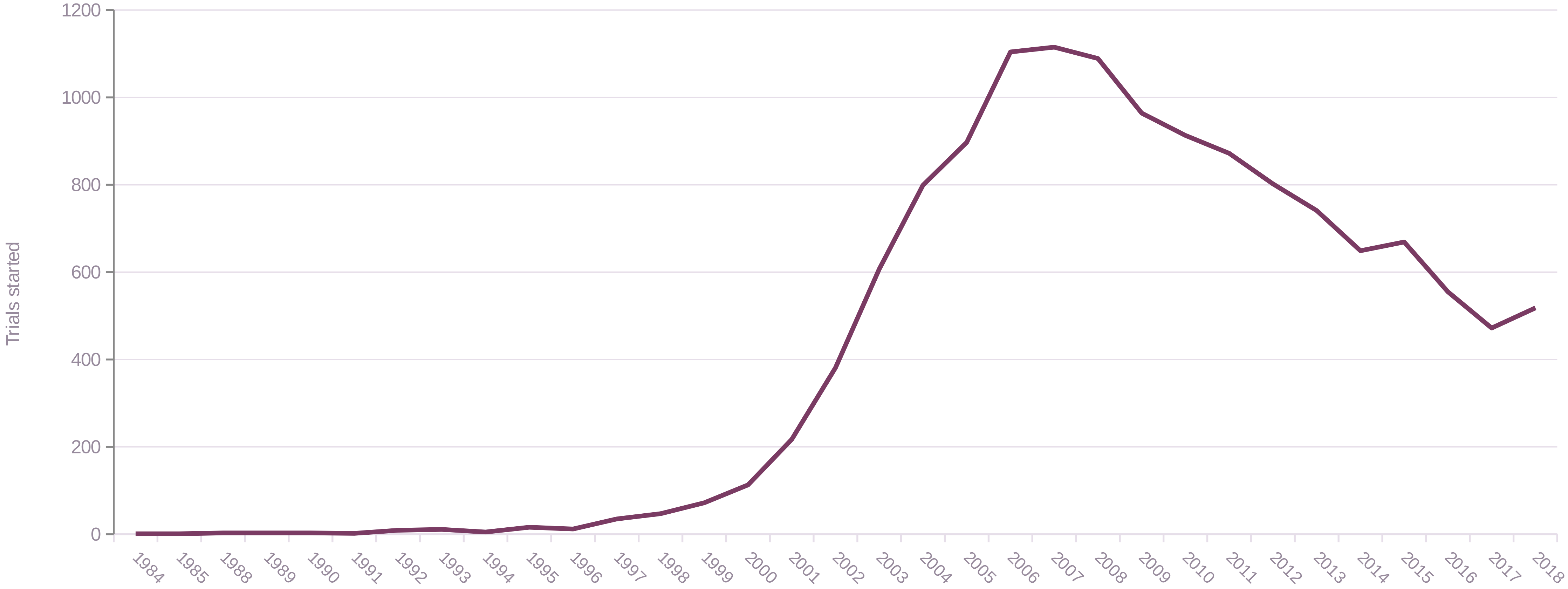
4

2019–2020 flagged, not hidden

Trial counts fall off a cliff in 2019–2020. This reads as a dataset extraction cutoff, not a real slowdown — flagged directly on the chart rather than silently shown as a trend.

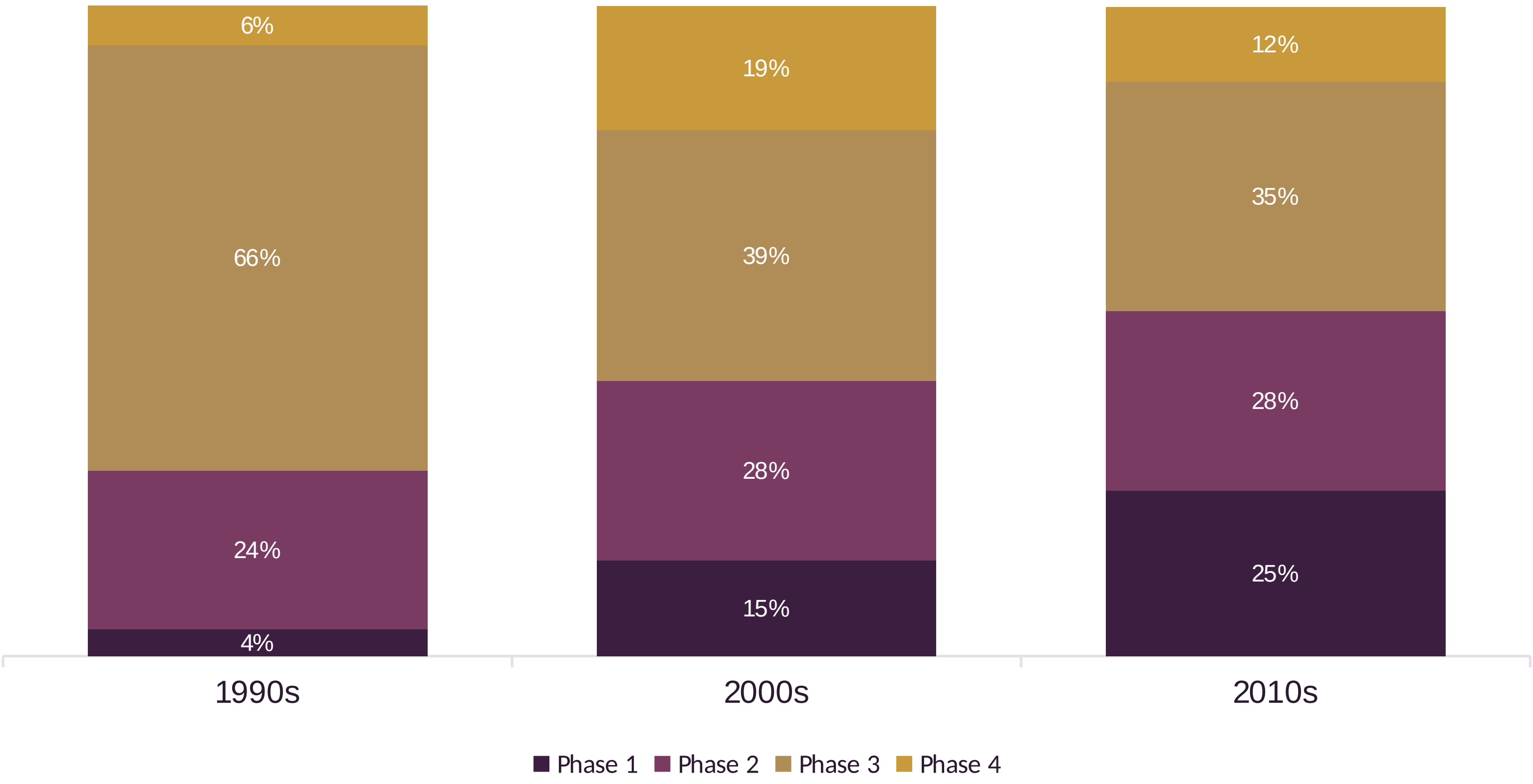
TRENDS OVER TIME

Trial launches grew nearly 400× from 1990 to a 2006–’08 peak, then gradually declined



1984–1989 and 2019–2020 omitted — too few trials per year to plot meaningfully alongside the core trend; see methodology.

Early-phase trials nearly tripled their share since the 1990s, as Phase 3's dominance faded



Phase 1 share

4.1% → 25.4%

1990s to 2010s

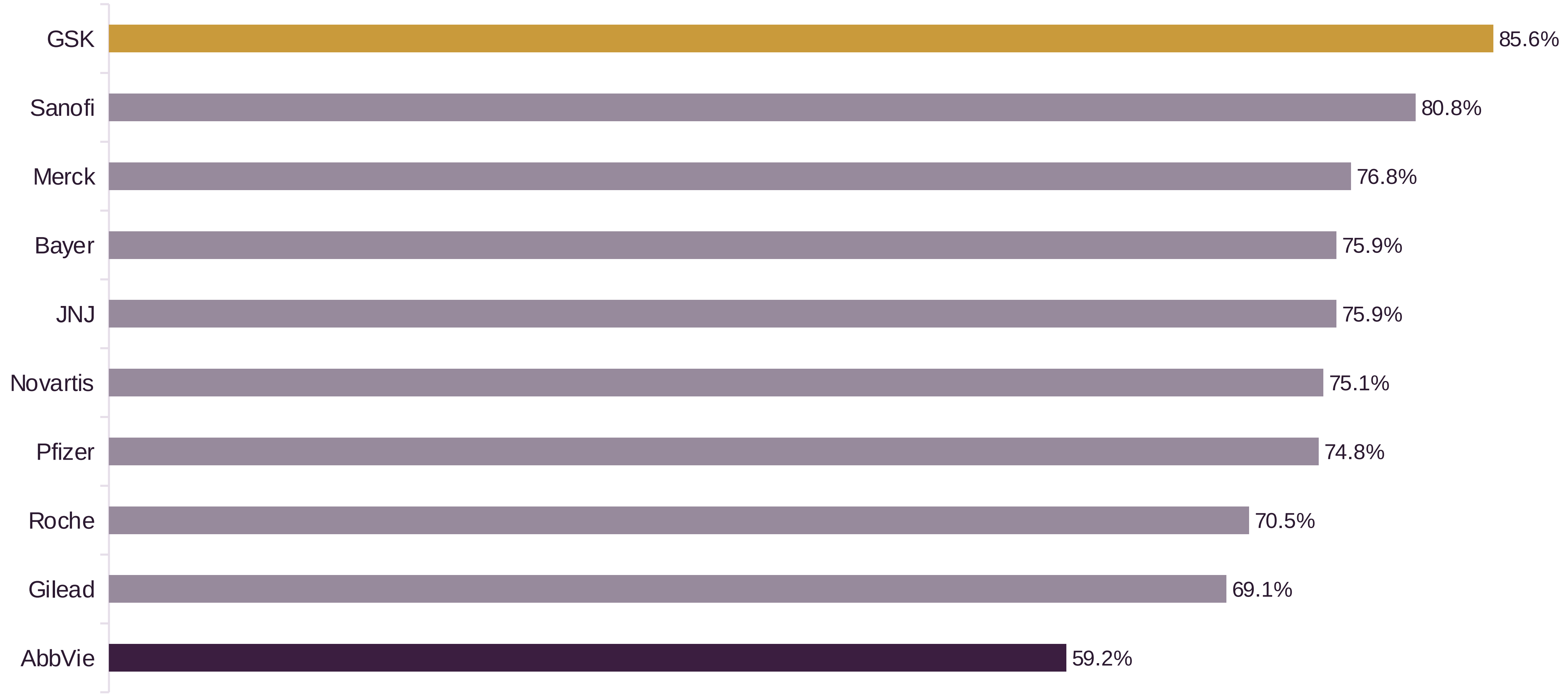
Phase 3 share

65.5% → 35.3%

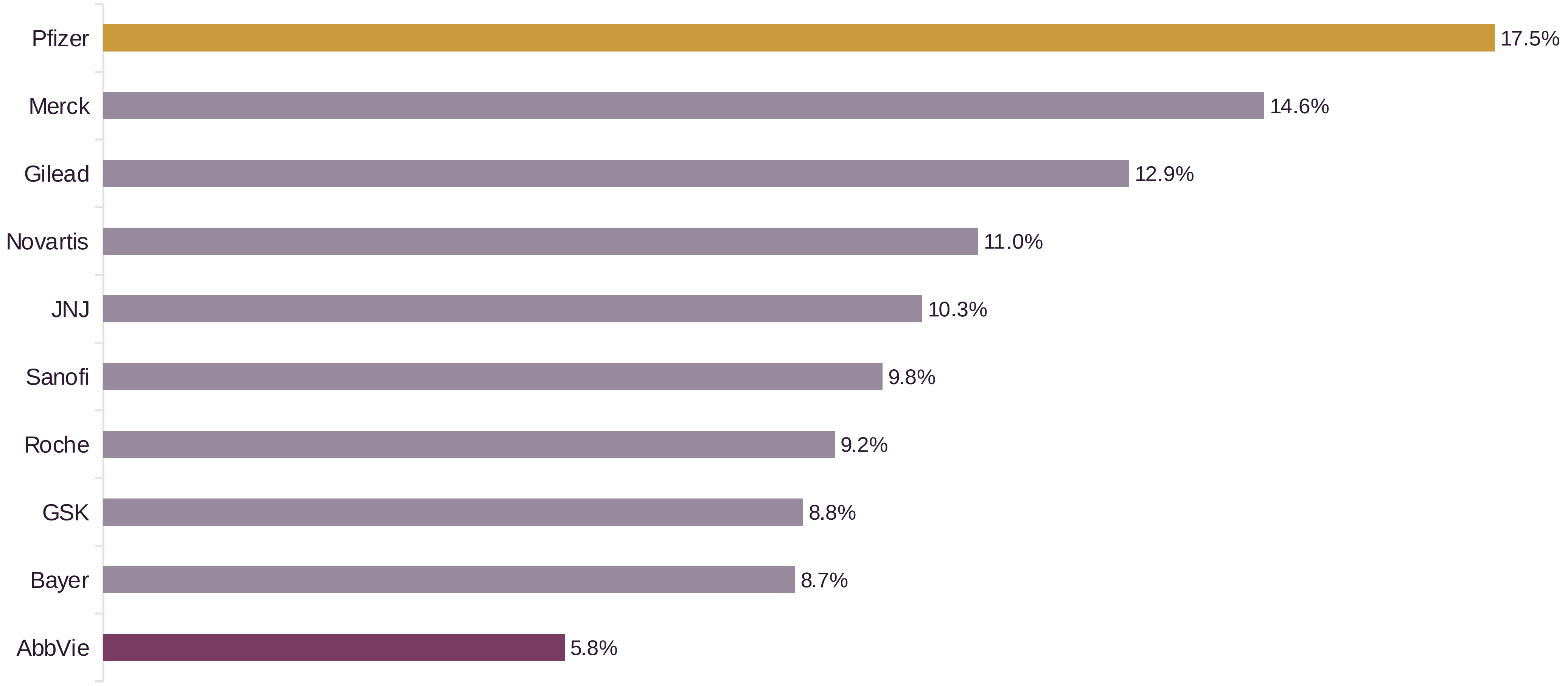
1990s to 2010s

Sponsors are testing more candidates earlier and cheaper, rather than betting everything on fewer Phase 3 mega-trials.

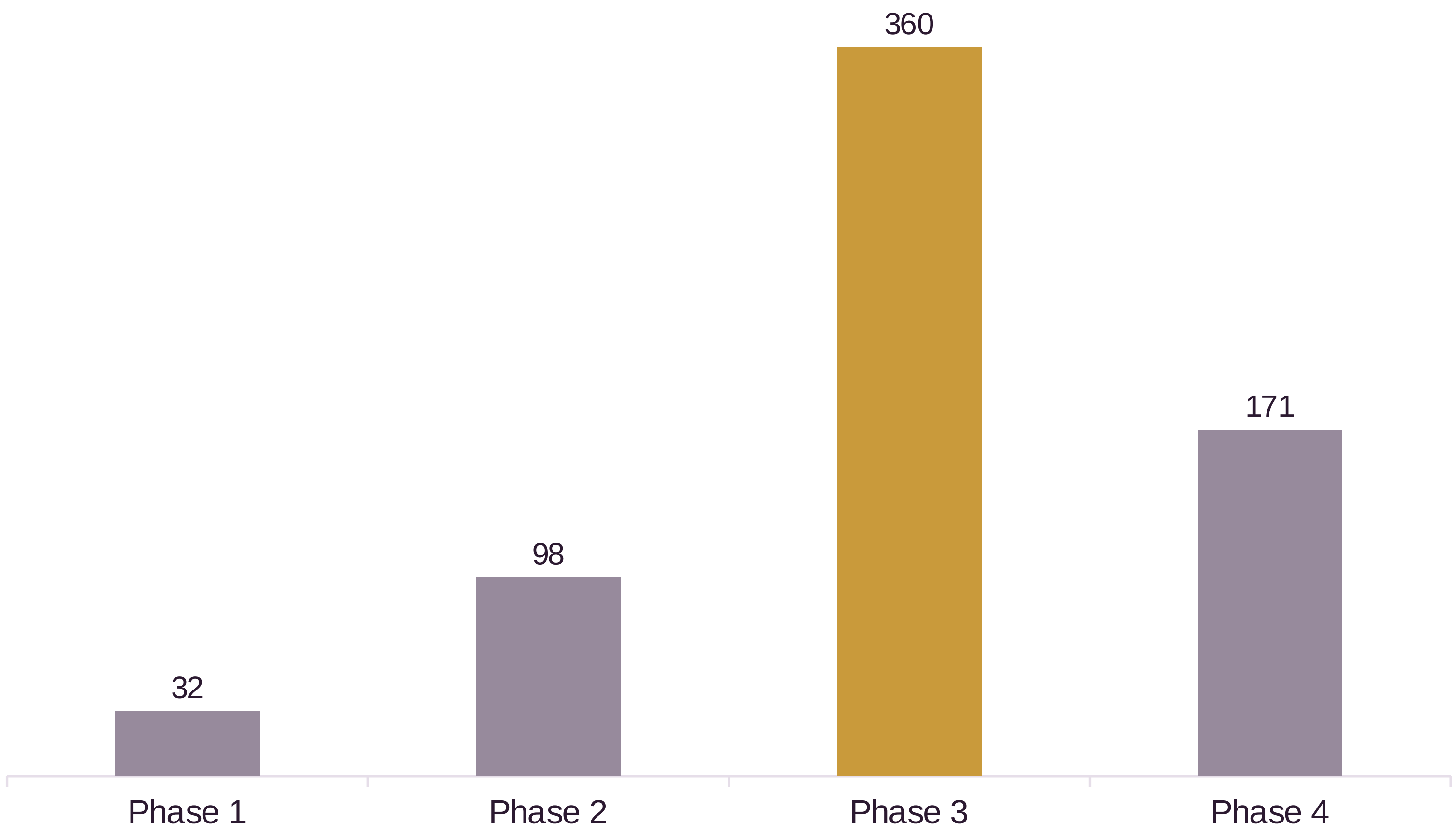
GSK completes 85.6% of its trials — 26 points ahead of AbbVie, the lowest of the ten



Pfizer halts nearly 1 in 5 trials before completion — three times AbbVie's rate



Phase 3 trials enroll a median of 360 patients — 11× Phase 1’s 32



Why it matters

Phase 1 tests safety in a small cohort. Phase 3 has to prove efficacy at population scale before regulators will sign off — the enrollment curve is really the cost curve of de-risking a drug.

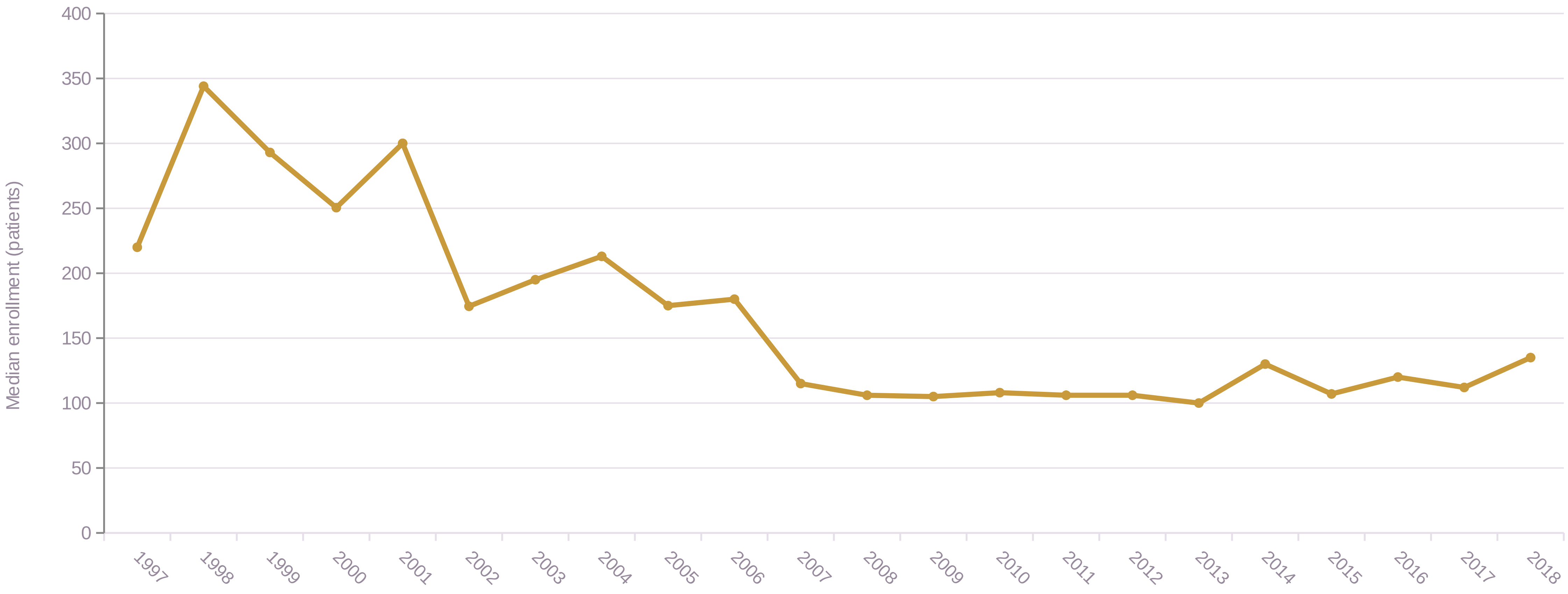
This is exactly the cost AI-assisted patient matching and adaptive trial design are aimed at compressing.

Gilead's pipeline is 62% infectious disease — the most concentrated focus of any major sponsor

Sponsor	Leading therapeutic area	Share of sponsor's trials
Gilead	Infectious Disease	62.2%
Roche	Oncology	52.2%
Bayer	Oncology	49.3%
GSK	Infectious Disease	**37.3%**
Novartis	Oncology	**33.8%**
AbbVie	Oncology	**33.7%**
JNJ	Neurological/Psychiatric	**33.7%**
Sanofi	Oncology	**33.7%**
Pfizer	Oncology	**30.1%**
Merck	Infectious Disease	**25.1%**

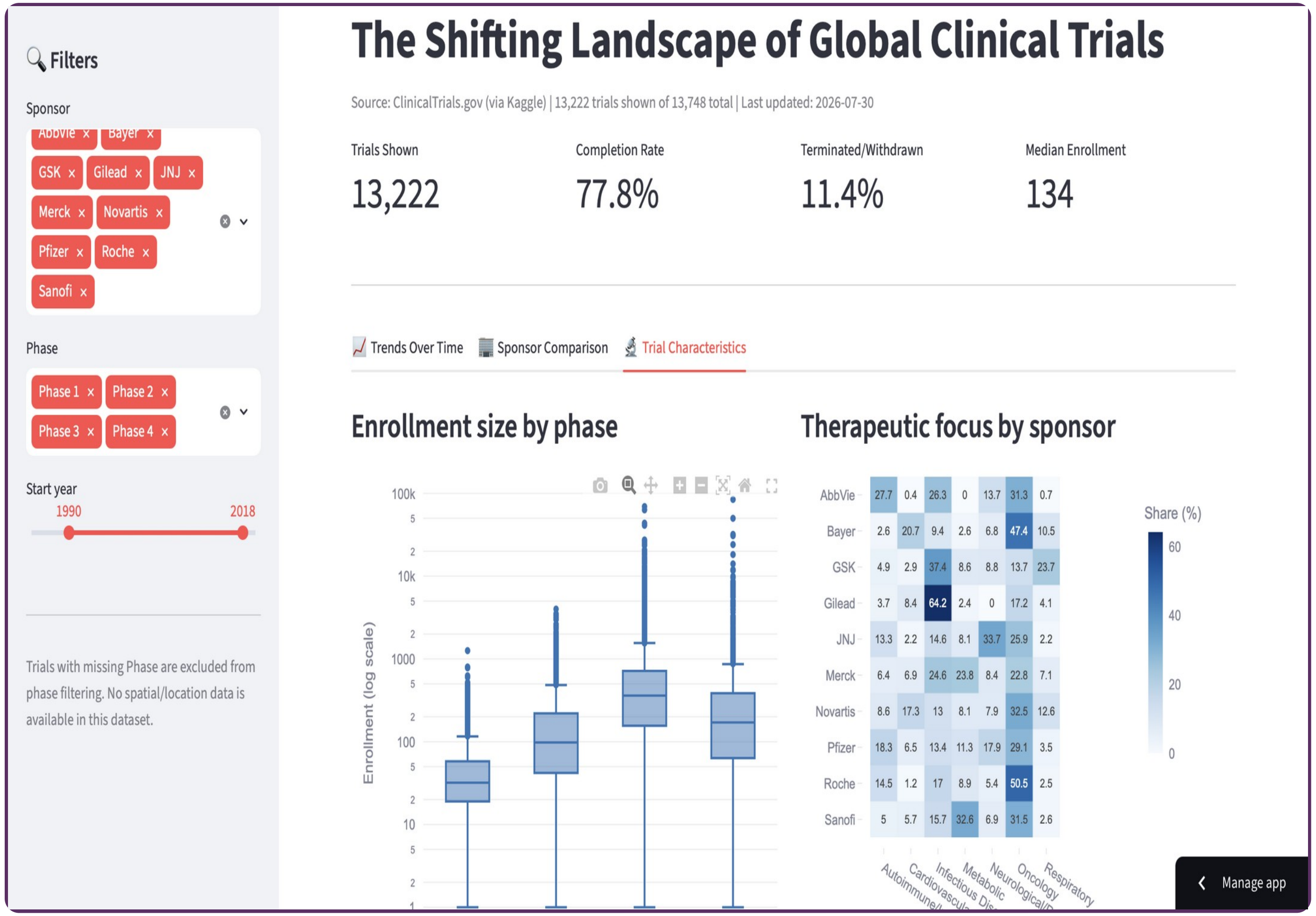
Rows with a leading specialty above 45% of the sponsor's trial volume are highlighted — a strong single-therapeutic-area bet, not a diversified portfolio.

Median trial size has nearly halved since the late 1990s — from ~300 to ~120 patients



Restricted to 1997–2018, where each year has ≥35 trials — enough for a stable median. Trials with Enrollment = 0 excluded.

Explore it yourself: a live, filterable dashboard



Three tabs, one shared filter set

Trends Over Time

Trial volume by year, phase mix by decade

Sponsor Comparison

Completion & termination rates, side by side

Trial Characteristics

Enrollment by phase, therapeutic-area heatmap

Sponsor, phase, and year filters apply live across all three tabs — not a static chart dump.

Live dashboard: siris-clinical-trials.streamlit.app

Source: github.com/shirisha-shashidhar/clinical-trials-final-project

SO WHAT

An industry becoming more cautious, more fragmented — and ready to be reshaped by AI

1. Trials are getting smaller and earlier-phase-heavy — sponsors are de-risking earlier rather than betting big on fewer Phase 3s.

2. Completion and termination rates vary 3× across sponsors doing the same kind of work — execution discipline, not just science, drives outcomes.

3. Sponsors are specializing (Gilead in infectious disease, Roche in oncology) rather than running generalist pipelines.

4. All of this is the pre-AI baseline. Digital health promises faster patient matching, adaptive designs, and leaner trials — the metrics on this deck are exactly what that promise needs to beat.

THANK YOU

The Shifting Landscape of Global Clinical Trials

GitHub repository

github.com/shirisha-shashidhar/clinical-trials-final-project

Live dashboard

siris-clinical-trials.streamlit.app